Soichiro Nishimori

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https://nissymori.github.io/

Research Summary

RL enthusiast, who is interested in scaling agents across computation, interaction, and data. Toward this goal, I design high-throughput JAX-based environments, exploration-oriented policy objectives, and robust algorithms for imperfect supervision, including noisy preferences and uncurated data.

Education

Apr. 2024 – Present	**Doctoral Program in Engineering, Graduate School of Frontier Sciences, The University of Tokyo, Chiba, Japan.** Supervisor: Professor Masashi Sugiyama. Theme: *Reinforcement Learning under Weak Supervision.*
Oct. 2025 – Feb. 2026	**Visiting Ph.D. Student, School of Computer Science, The University of Sydney, Sydney, Australia.** Supervisor: Professor Tongliang Liu.
Apr. 2022 – Mar. 2024	**Master of Engineering, Graduate School of Frontier Sciences, The University of Tokyo, Chiba, Japan.** Supervisor: Professor Masashi Sugiyama. Theme: *Offline Reinforcement Learning under Weak Supervision.*
Apr. 2018 – Mar. 2022	**Bachelor of Integrated Human Studies, Kyoto University, Kyoto, Japan.** Supervisor: Professor Shin Ishii. Theme: *Partially Observable Markov Decision Process (POMDP).*

Research Publications

International Conference

1. **S. Nishimori** et al., "Emergence of exploration in policy gradient reinforcement learning via retrying," in *ICML*, 2026. URL: `https://arxiv.org/abs/2606.00151`,
   Proposed **ReMax**, a policy optimization objective that encourages exploration in policy-gradient RL by optimizing the best return over multiple trials.
2. **S. Nishimori**, X.-Q. Cai, J. Ackermann, and M. Sugiyama, "Offline reinforcement learning with domain-unlabeled data," in *RLC*, 2025. URL: `https://arxiv.org/abs/2404.07465`.
3. Y. Tang, Y. Zhang, J. Ackermann, Y.-J. Zhang, **S. Nishimori**, and M. Sugiyama, "Recursive reward aggregation," in *RLC*, 2025. URL: `https://arxiv.org/abs/2507.08537`.
4. S. Koyamada, **S. Nishimori**, and S. Ishii, "A batch sequential halving algorithm without performance degradation," in *RLC*, 2024. URL: `https://arxiv.org/abs/2406.00424`.
5. S. Koyamada et al., "PGX: Hardware-accelerated parallel game simulators for reinforcement learning," in *NeurIPS*, 2023. URL: `https://arxiv.org/abs/2303.17503`,
   Contributed to the Backgammon environment in PGX, a JAX-based framework for hardware-accelerated board-game simulation.
6. S. Koyamada, K. Habara, N. Goto, S. Okano, **S. Nishimori**, and S. Ishii, "MJX: A framework for Mahjong AI research," in *2022 IEEE Conference on Games (CoG)*, 2022. URL: `https://ieee-cog.org/2022/assets/papers/paper_162.pdf`.

Journal

1. **S. Nishimori**, Y.-J. Zhang, T. Lodkaew, and M. Sugiyama, "On symmetric losses for policy optimization with noisy preferences," *Transactions on Machine Learning Research (TMLR)*, 2026. URL: `https://openreview.net/forum?id=cBWGLmSeao`.

Projects

1. **S. Nishimori**, S. Okano, K. Habara, S. Koyamada, E. Yu, and M. Sugiyama, *Mahjax: A GPU-accelerated mahjong simulator for reinforcement learning in JAX.* 2025. URL: `https://github.com/nissymori/mahjax`,
JAX-based Riichi Mahjong environment for pure self-play RL, reaching over **1M steps/second**.
2. **S. Nishimori**, *JAX-CORL: Clean single-file implementations of offline reinforcement learning algorithms in JAX.* 2024. URL: `https://github.com/nissymori/JAX-CORL`.

Preprints

1. **S. Nishimori**, S. Okano, K. Habara, S. Koyamada, E. Yu, and M. Sugiyama, "Mahjax: A GPU-accelerated mahjong simulator for reinforcement learning in JAX," *arXiv preprint arXiv:2605.20577*, 2026. URL: `https://arxiv.org/abs/2605.20577`.
2. **S. Nishimori** and P. Parmas, "Retry policy gradients in continuous action spaces," *arXiv preprint arXiv:2606.05888*, 2026. URL: `https://arxiv.org/abs/2606.05888`.
3. S. Ono, J. Ackermann, **S. Nishimori**, T. Ishida, and M. Sugiyama, "Mitigating reward hacking in RLHF via advantage sign robustness," *arXiv preprint arXiv:2604.02986*, 2026. URL: `https://arxiv.org/abs/2604.02986`.
4. P. Parmas et al., "Ordergrad: Optimizing beyond the mean with order-statistic policy gradient estimation," *arXiv preprint arXiv:2606.06096*, 2026. URL: `https://arxiv.org/abs/2606.06096`.
5. S. Takashiro*, **S. Nishimori***, P. Parmas*, et al., "On advantage estimates for max@k policy gradients," *arXiv preprint arXiv:2606.06080*, 2026, *Equal contribution. URL: `https://arxiv.org/abs/2606.06080`.
6. B. Tong, J. Komiyama, **S. Nishimori**, and P. Parmas, "Finite-time regret analysis of retry-aware bandits," *arXiv preprint arXiv:2605.20854*, 2026. URL: `https://arxiv.org/abs/2605.20854`.
7. T. Kitamura et al., "A policy gradient primal-dual algorithm for constrained MDPs with uniform PAC guarantees," *arXiv preprint arXiv:2401.17780*, 2024. URL: `https://arxiv.org/abs/2401.17780`.

Experience

Jul. 2022 – Sep. 2023 — **Research Intern**
OMRON SINICX – Tokyo, Japan

Apr. 2024 – Present — **Part-time Researcher**
RIKEN AIP – Tokyo, Japan

Aug. 2024 – Oct. 2024 — **Research Intern**
Matsuo-Iwasawa Lab, The University of Tokyo – Tokyo, Japan

Fellowships and Grants

Apr. 2024 – Mar. 2027 — **Research Fellowship for Young Scientists (DC1)**
Japan Society for the Promotion of Science (JSPS)
JPY 200,000/month; JPY 2,200,000 over three years as research funding

Fellowships and Grants (continued)

Aug. 2025 — **Travel Grant**
Reinforcement Learning Conference (RLC) 2025
USD 825

Skills

Languages	Japanese, English (TOEFL iBT 96).
Programming	Python, **JAX**, PyTorch, and LaTeX.